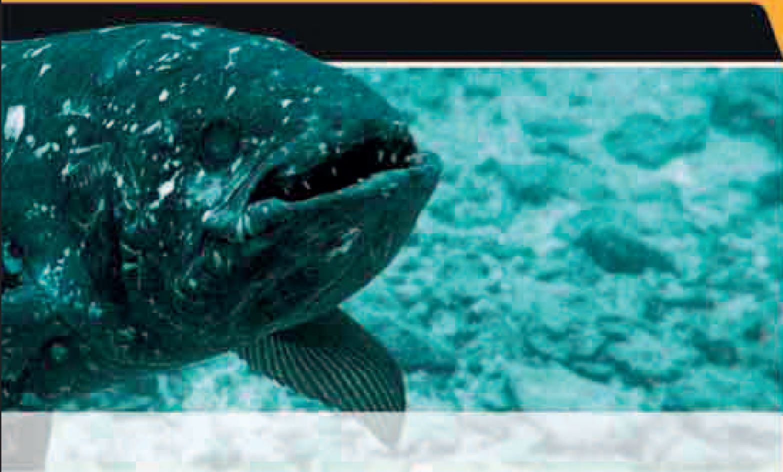
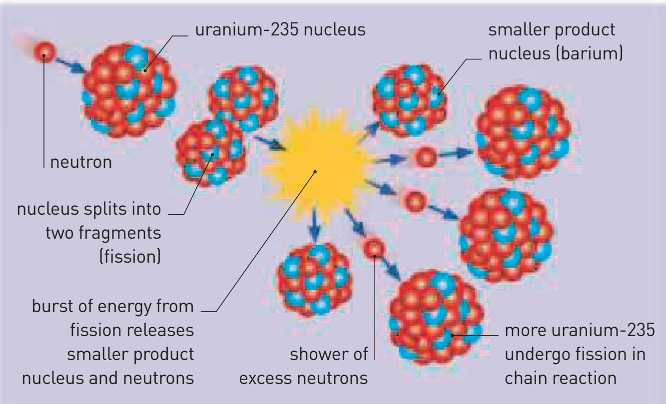
1939



Made by German company Heinkel in 1939, the first jet aircraft—the He178—was the prototype for later models built for combat at the end of World War II.



NUCLEAR FISSION CHAIN REACTION

Nuclear fission releases atomic energy—which is maximized when the process is done in a way that initiates a chain reaction. Uranium-235 is favored as it is the most abundant fissionable isotope and has plenty of critical neutrons. When the uranium nucleus (fissionable nucleus) is bombarded with an external source of neutrons, the uranium nuclei split into smaller fragments. More neutrons are emitted as by-products, which in turn split other uranium nuclei in the chain reaction.

(1907–2004), curator at a museum in East London, South Africa, was asked to **collect a specimen** from the local fishing docks. There she found an unusual fish that she could not identify and, in the absence of any technical support, reluctantly had it stuffed. South African zoologist **James Smith** (1897–1968) later **identified it as a coelacanth**—a fish until then known only from fossils and **thought to have become extinct** with the dinosaurs. The coelacanth belongs to an **ancient group of lobe-finned fishes** related to the ancestors of the first vertebrates that evolved to live on land. Initially, the modern coelacanth was known only in deep waters of the western Indian Ocean, but a second species was found in Indonesian waters in 1997.

IN APRIL, GROTE REBER DISCOVERED a new kind of astronomical object with his radio telescope—**a radio galaxy called Cygnus A**. This object remains one of the strongest sources of radio signals that has ever been detected.

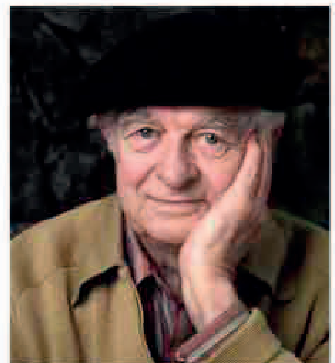
Six years after Hungarian-born physicist Léo Szilárd (1898–1964) conceived the idea of releasing energy in a nuclear chain reaction, there was **growing concern among scientists** that Nazi Germany would **develop an atomic bomb**. Italian physicist Enrico Fermi and German chemist Otto Hahn had already demonstrated that controlled nuclear fission by a chain reaction was possible. In August, German-born physicist **Albert Einstein**, by now in exile in America and working at Princeton University, wrote to **President Roosevelt** expressing the scientists' concerns. He later urged the US president to enter the race to be the first to make the atomic bomb. Einstein had prompted what would become the **Manhattan Project**: the Allied program of research and development that would eventually unleash the only **nuclear weapons so far used in wartime combat**.

Although many scientists left Germany in the months leading up to World War II, others stayed behind to continue their work in advancing science and technology. The first **jet engine was designed** and patented by German physicist **Hans von Ohain** (1911–98). On August 27, the first aircraft to fly under turbojet power—the **Heinkel He178**—had its **maiden flight**. After the war, von Ohain was one of many German scientists recruited to help advance scientific research in America as part of **Operation Paperclip**.

American chemist **Linus Pauling** published his most celebrated book: *The Nature of the Chemical Bond*. In it he described **the behavior of electrons** in forming ionic bonds (different atoms bonding by gaining and losing electrons) and covalent bonds (atoms bonding by sharing electrons). In particular, Pauling developed the idea that shared electrons are not in a fixed position, but **orbit around both nuclei associated with the bond**.

> “**SCIENCE** IS THE **SEARCH FOR TRUTH**—IT IS **NOT A GAME** IN WHICH ONE TRIES TO **BEAT HIS OPPONENT** OR DO **HARM** TO **OTHERS.**”

Linus Pauling, American chemist and peace activist, in *Liberation,* 1958



LINUS PAULING
(1901–94)

American chemist and peace activist, Linus Pauling is the only person to have received two unshared Nobel Prizes. His work spanned quantum mechanics in chemistry and the structures of complex biological molecules, such as proteins. After World War II he campaigned against the further use of nuclear weapons.

September 15 German physicist **Carl von Weizsäcker** describes the energy-releasing fusion reactions that happen in a star

October 27 Du Pont Company announces that polyamide 6-6 would be called **nylon**

December 17 Otto Hahn reports carrying out **nuclear fission**

December 23 A live **coelacanth** is found off the coast of **South Africa**

February 11 Swedish and German physicists **Lise Meitner** and **Otto Frisch** explain the scientific basis of the **nuclear fission** reported by Otto Hahn

February American and Canadian physicists **Robert Oppenheimer** and **George Volkoff** describe upper limit of mass of neutron star

April Grote Reber discovers radio galaxy Cygnus A

August 2 Albert Einstein writes to President Roosevelt about using uranium for an atomic bomb

October 11 Einstein writes to President Roosevelt again, urging the USA to develop the atomic bomb

December American biologists **Karl Hamner** and **James Bonner** explain how day and night length determine flowering in plants (**photoperiodism**)

Late 1939 Paul Müller discovers that DDT can be used as a contact poison to kill insects